

MODULE 1

BASICS 8086 ARCHITECTURE

- **The microprocessor is a programmable integrated device that has computing and decision making capability**
- A microprocessor is an IC chip which can act as CPU of a digital computer
- The microprocessor communicates and operates in the binary numbers 0 and 1
- Microprocessor reads binary instruction from a storage device called memory,
- Accepts binary data as input and processes data according to those instruction, and provides results as output.

Binary Digits

- The microprocessor operates in binary digits 0 and 1 known as bits
- These digits 0 and 1 are also synonymous with low and high voltages
- Each microprocessor recognizes and processes a group of bits called the word
- Microprocessors are classified according to their word length.
(for example a processor with an 8-bit word is known as an 8 bit microprocessor, and a processor with a 32-bit word is known as a 32-bit microprocessor)

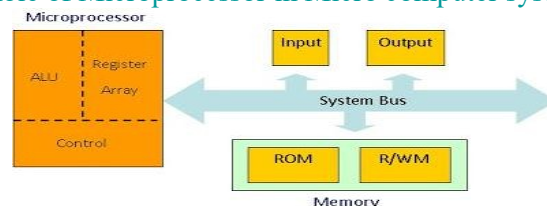
Evolution of Intel Microprocessor

- In 1971, the first microprocessor Intel 4004 was invented,.It has word length of 4 bits
- Then later 8 bit microprocessor was developed by Intel in 1972 and it was named as 8008(It got lot of performance issues)
- The upgraded version of 8008 ,8080 was introduced in 1974 (Its main drawback was this that it required three power supplies)
- The modified version of 8080 is 8085.It has a word length of 8 bits.
- In 1979, intel developed microprocessor 8086 which is a 16 bit microprocessor.
After the development of 8086 a number of microprocessors have been developed by various companies such as 8088, 80186, 80188, 80386, 80486, 80487, Pentium 1 to Pentium 5 etc.
- **Memory:** Memory is a medium that stores binary information called instructions and data.
- Provides the instructions and data to microprocessor.
- Stores results and data for the microprocessor.

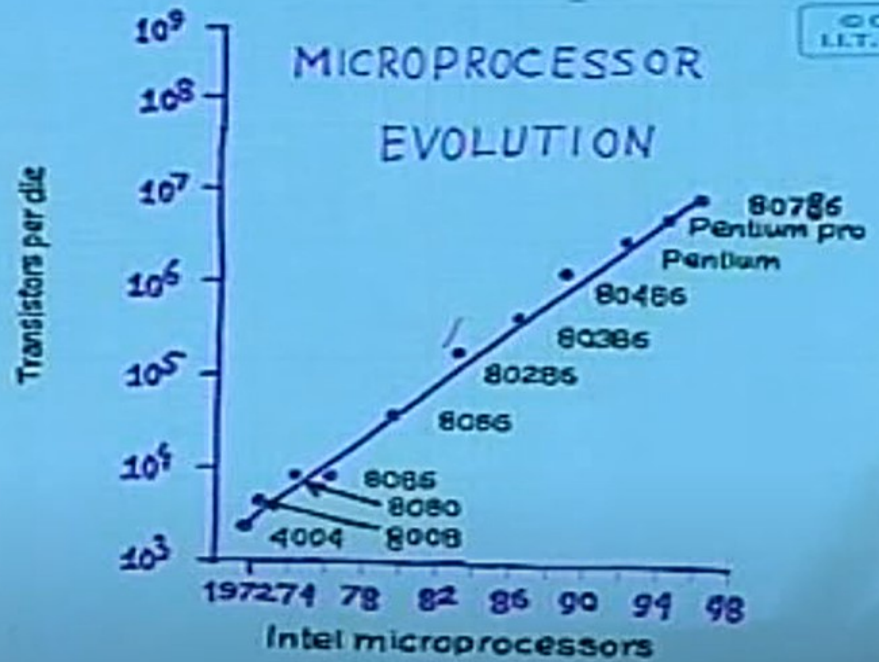
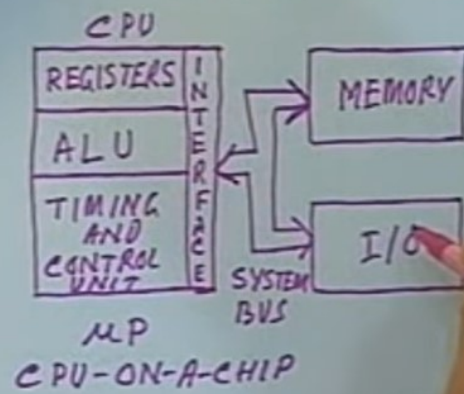
Input/Output Device: An input device is the one which transfers information from outside world to the computer.

- Examples:Keyboard,mouse
- An output device is the one which transfers information from the computer to the outside world.
Printer,Monitor,Speaker,Plotter etc

Role of Microprocessor in Micro computer system

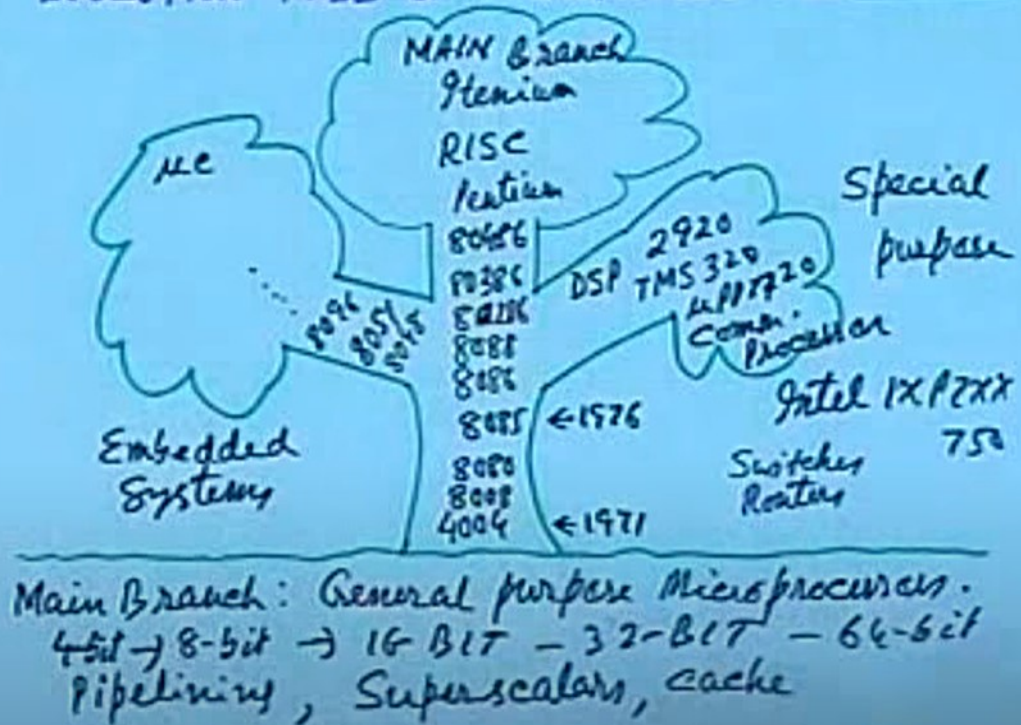


MICROPROCESSOR ARCHITECTURE



EVOLUTION TREE OF MICROPROCESSORS

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FEATURES OF MICROPROCESSORS

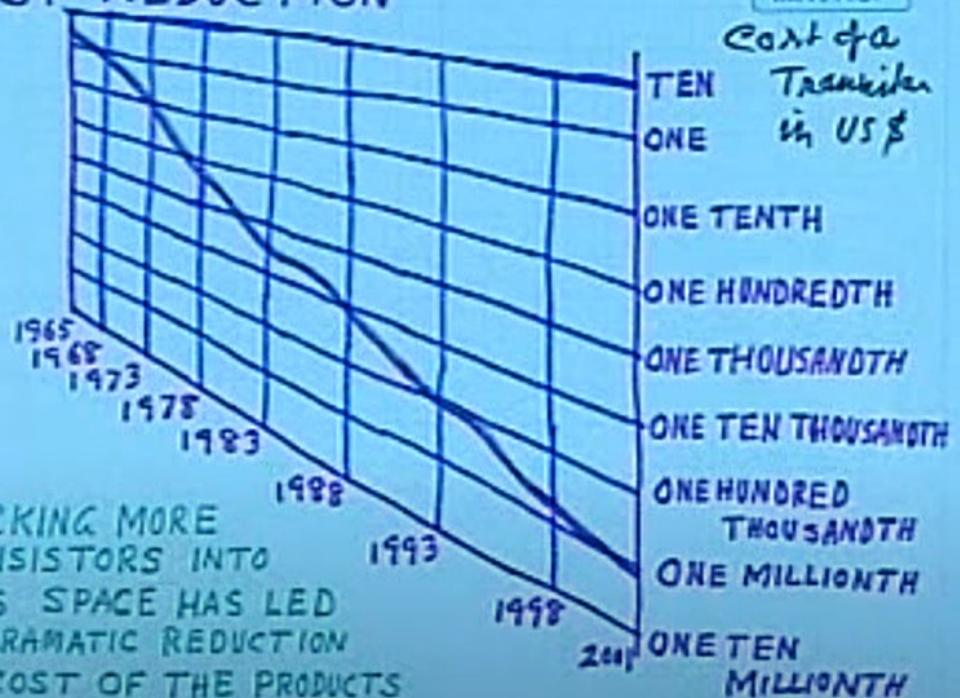
• Typical Features:

- Smaller size
- Lower cost
- Higher reliability
- Higher flexibility and versatility
- Functionally more powerful
- Lower power consumption

COST REDUCTION

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Cost of a
Transistor
in US\$



PACKING MORE
TRANSISTORS INTO
LESS SPACE HAS LED
TO DRAMATIC REDUCTION
IN COST OF THE PRODUCTS
THEY POPULATE.

MOORE'S LAW

OCET
I.I.T. KGP

PROCESSOR	YEAR OF INTRODUCTION	# OF TRANSISTORS
4004	1971	2250
8008	1972	2500
8080	1974	5000
8086	1978	29000
i286	1982	120000
i386	1985	275000
i486	1989	1,180,000
Pentium	1993	3,100,000
Pentium II	1997	7,500,000
Pentium III	1999	24,000,000
Pentium IV	2000	42,000,000

APPLICATION OF MICROPROCESSORS

General Purpose:

- Use in PCs, Laptops, Workstations, Super-minis and Supercomputers.

Embedded Applications:

- Consumer Electronics
- Instrumentation
- Process control
- Communications
- Office
- Emerging multimedia applications
- Process Controls

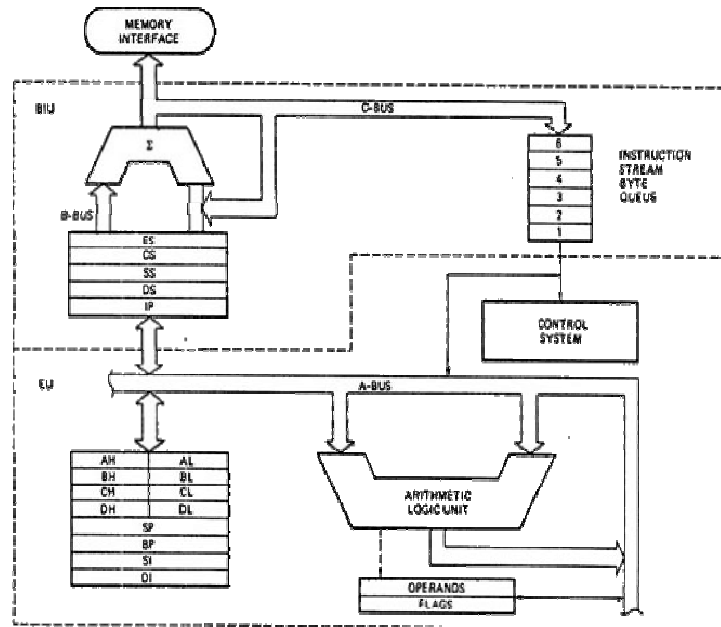
- **MICROPROCESSOR:** A Microprocessor is a semiconductor electronic clock driven device manufactured by using LSI/VLSI technique
- The microprocessor unit reads instructions from the memory and perform the tasks specified.
- It communicates with input/output devices either to accept or it send data.
- The timing of the communication process is controlled by the group of circuits called the control unit.

8086-Details

- It is a 16 bit microprocessor.
- It operates on +5 volts .
- It is available in 40 pin DIP
- It provides a 16-bit ALU.

- The size of address bus is 20-Bit Bus, i.e. it can access up to 1 MB of Memory-(2²⁰=1,048,576 bytes)
- It supports two operating modes- maximum mode and minimum mode
- It operates in 5 MHz/8MHz/10MHz/12MHz clock frequency
- Memory is divided into two banks-even bank and odd bank.
- Address range of memory is from 0000H to FFFFH.

ARCHITECTURE of 8086



8086 Microprocessor is divided into two units. They are

- **Bus Interface Unit (BIU)**
- **Execution unit (EU)**

Bus interface unit(BIU)

- The functions of BIU is that it fetches instructions
- Reads data from memory or I/O ports
- Write data to memory or I/O ports and buffers them in a queue
- Queue is 6 bytes long, it is referred to as instruction pipe line
- BIU fills this space with instructions waiting for execution by the EU
- **The BIU fetches and stores instructions in the queue while the EU executes present instructions, This is called pipelining.**

Register Organisation of 8086

The intel 8086 contains fourteen 16-bit registers. They are grouped as :-

- 1) General Purpose Registers
- 2) Pointer & Index Registers
- 3) Segment Registers
- 4) Instruction Pointer and Flags

1.General Purpose Register(GPRs):

- Intel 8086 has four 16 bit GPRs ,Each is further divided in to two 8-bit registers (higher order and lower order)
- 1) Accumulator AX- AH,AL
- 2) Base Register BX-BH,BL
- 3) Counter Register CX-CH,CL

2. Pointer & Index Register

- They are used for computing the memory address ,These registers are 4 namely
 - 1) Stack Pointer (SP)
 - 2) Base Pointer (BP)
 - 3) Source Index (SI)
 - 4) Destination Index (DI)

3. Segment Registers: There are 4 namely

- 1) Code Segment Register (CS)-Used to store instructions of a program
- 2) Data Segment Register (DS)-Store program data,variables & constants
- 3) Stack Segment Register (SS)-If stack is used then SS is used
- 4) Extra Segment Register (ES)-Stores some data strings

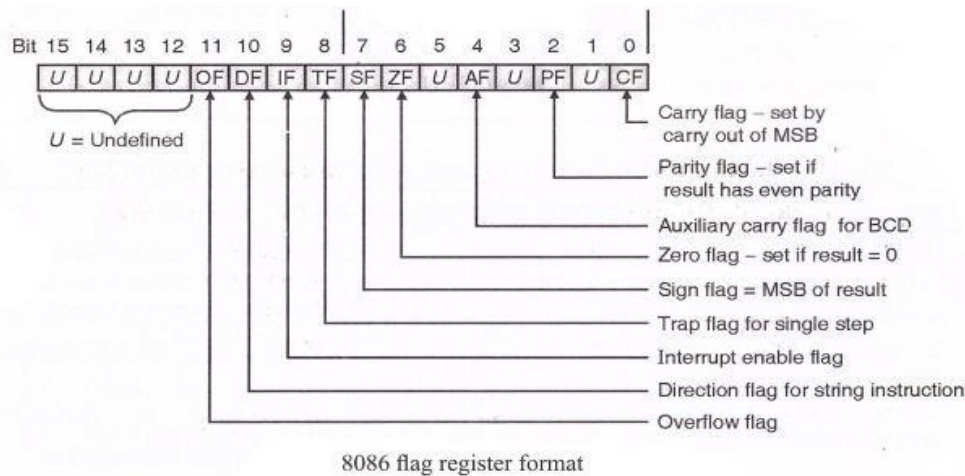
4. Instruction Pointer (IP) & Flag

- The IP points to the next instruction in current code segment.
- 8086 has 9 flag which are subdivided in to status & control flags.
 - ▶ **ALU:** It performs 16 bit or 8bit arithmetic operations like addition, subtraction ,multiplication, division, data conversion and logical operations like NOT,OR,AND,XOR.
 - ▶ It also performs register increment, decrement and shift operations.

Execution Unit: The functions of execution unit are:

- 1) To tell BIU where to fetch the instructions or data from.
- 2) To decode the instructions.
- 3) To execute the instructions.
- 4) EU has 16-bit ALU, which can perform arithmetic and logical operations on 8-bit as well as 16-bit.
- 5) EU contains the GPRs, pointer and index registers and flags.

Flag register (Status register)



Conditional flags/Status flags:

There are six conditional flags. These flags are set or reset by the EU on the basis of the result of the arithmetic and logical operation.

They are :-

i. **Carry flag** – This flag is set, if there is a carry in case of addition or borrow in case of subtraction

ii. **Parity flag (Pf)** – This flag is set, if the result contains even number of 1s

iii. **Auxiliary Carry Flag:** This flag is used for BCD operation.

iv. **Zero Flag:** This flag is set, if the result of arithmetic or logical operations is zero

v. **Sign flag:** it is set if arithmetic operation causes negative results.

vi. **Overflow flag** - This flag sets if an overflow occurs.

This flag sets to 1 if the result of a signed operation is too large to fit in the number of bits available to represent it.

Control Flags: There are 3 control flags in 8086 processor:-

- 1) Trap flag
 - 2) Interrupt Enable flag
 - 3) Direction flag
- **Trap flag** - it permits the operation of a processor in single-step mode
 - **Interrupt flag** - if the flag is set maskable hardware interrupts will be enabled, if set to 0, such interrupts will be ignored.
 - **Direction flag** – This is used by string manipulation instructions.
 - *If this flag is set, the contents of index registers (SI and DI) will be decremented after each operation*

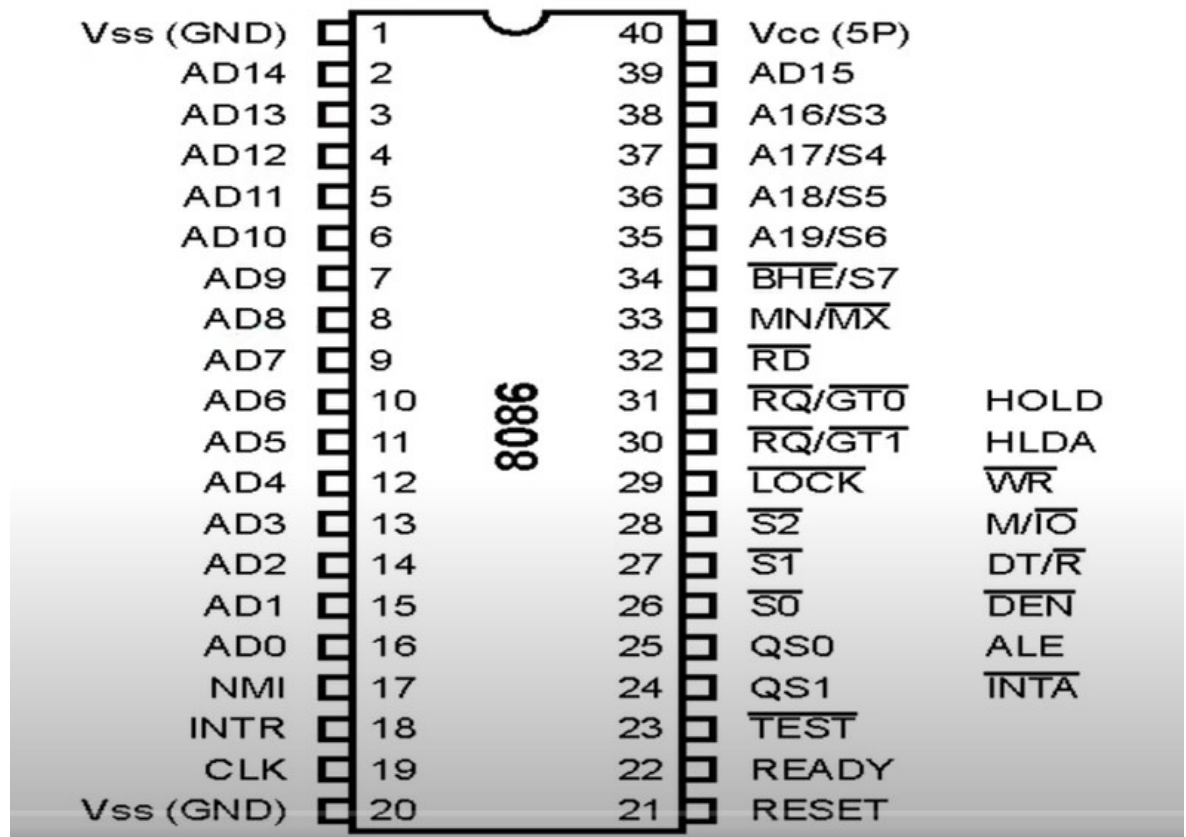
PIN Diagram of 8086 (HARDWARE ARCHITECTURE)

- 8086 available in 40 pin DIP
 - 8086 operates in minimum mode (single processor) or maximum mode (multi processor)
 - Functions of some pins are different in minimum mode and maximum mode
- Thus the 8086 signals are categorized into 3 groups

- i. The signals having common functions in both minimum and maximum mode
- ii. The signals that have special functions for minimum mode

- iii. The signals having special functions for maximum mode

8086 Pin description



Categorized in to three groups:

1. Signals for minimum mode only. ✓
2. Signals for maximum mode only. ✓
3. Signals common to both minimum and maximum mode. ✓

Minimum mode only

Name	Function
HOLD	Hold request
HLDA	Hold acknowledge
WR	Write control
IO/M	IO/memory control
DT/R	Data transmit/receive
DEN	Data enable
SSO	Status line
ALE	Address latch enable
INTA	Interrupt acknowledge

Maximum mode only

Name	Function
RQ/GT1, 0	Request/grant bus access control
LOCK	Bus priority lock control
S2-S0	Bus cycle status
QS1, QS0	Instruction queue status

Common to both minimum & maximum mode:

AD7-AD0	Address/data bus
A15-A8	Address bus
A19/S6-A16/S3	Address/status
MN/ $\overline{\text{MX}}$	Minimum/maximum Mode control
$\overline{\text{RD}}$	Read control
$\overline{\text{TEST}}$	Wait on test control
READY	Wait state control
RESET	System reset
NMI	Nonmaskable Interrupt request
INTR	Interrupt request
CLK	System clock
V _{CC}	+5 V
GND	Ground

- The 8086 works in two modes, minimum and maximum. Accordingly pin configuration changes. They are 32 signals common to both modes. Remaining 8 pins are different for each mode.
- Address/Data bus:** Carries 16-bit address/data. AD7-AD0 carries lower byte data/address. AD15-AD8 carries higher byte data/address.
- When **ALE (Address Latch Enable)** is high, address/data bus carries address. Otherwise it carries data.
- A16/S3- A19/S6: Address/status bus.** When ALE is high, it carries address. Otherwise it carries status signal.

S4	S3	Function
0	0	Extra Segment
0	1	Stack Segment
1	0	Code Segment
1	1	Data Segment



- S5: indicated the presence of interrupt.
- S6: shows the status of bus master for current operation. When S6 is zero, indicates currently 8086 is holding the bus.
- **INTR: Interrupt Request**: used to request a hardware interrupt of INTR is held high when interrupt enable flag is set, the 8086 enters an interrupt acknowledgement cycle after the current instruction has completed its execution
- **NMI: Non-maskable interrupt**. Similar to INTR except that the NMI interrupt does not check for interrupt enable flag is at logic 1, i.e, NMI is not maskable internally by software.
- **Clock** : Clock signal is provided through this pin. It provides timing to the processor for operations. Its frequency is different for different versions, i.e. 5MHz, 8MHz and 10MHz.

(MORE DETAILS.....)

1. Signals common for min and max mode

- The 8086 require +5V (*VCC at pin 40*) and one clock pulse (pin 19) for operation
- Pin number 1 and 20 are grounded
- 20 pins are used as address lines.
- In this 20 pins, 16 pins are used as both address and data lines. (*multiplexing facility reduces the number of pins*).
- **Address/data bus - (Pin 2 to 16)**

AD0-AD15: Bidirectional Address/data Bus: These are lower order address bus multiplexed with data bus.

A16-A19/S3-S6: These are multiplexed. These are the 4 address/status buses.

- **S7/BHE - (Pin 34)**

BHE stands for Bus High Enable. It is used to indicate the transfer of data using data bus D8-D15

- **RD - (Pin 32)**

It is used to read signal for Read operation

- **Ready - (pin 22)**

When it is high, it indicates that the device is ready to transfer data. When it is low, it indicates wait state

- **RESET - (pin 21)**

It is used to restart the execution. It causes the processor to immediately terminate its present activity

- **INTR - (pin 18)**

It is an interrupt request signal

- **NMI - (pin 17)**

It stands for non-maskable interrupt.

- **MN/M - (pin 33)**

It stands for Minimum/Maximum. When it is high, it works in the minimum mode and vice-versa

S ₄	S ₃	Indications
0	0	Alternate Data
0	1	Stack
1	0	Code or None
1	1	Data

- **Minimum Mode Signals**

- INTA-(pin 24)

It is an interrupt acknowledgement signal. When the microprocessor receives this signal, it acknowledges the interrupt

- ALE - (pin 25)

It stands for address enable latch. This signal indicates the availability of a valid address on the address/data lines.

- DEN-(pin 26)

It stands for Data Enable. It is used to enable transreceivers to separate data from multiplexed AD Bus

- DT/R(pin 27)

It stands for Data Transmit/Receive signal. It decides the direction of data flow through the transceiver. When it is high, data is transmitted out and vice-a-versa.

- M/B(pin 28)

This signal is used to distinguish between memory and I/O operations. When it is high, it indicates I/O operation and when it is low indicates the memory operation.

- WR(pin 29)

It stands for write signal. It is used to write the data into the memory or the output device depending on the status of M/IO signal.

- HLDA-(pin 30)

It stands for Hold Acknowledgement signal. This signal acknowledges the HOLD signal.

- HOLD-(pin 31)

This signal indicates to the processor that external devices are requesting to access the address/data buses.

Maximum Mode Signals

- QS1 and QS0-(pin 24 and 25): These are queue status signals. These signals provide the status of instruction queue. Their conditions are shown in the following table

QS ₀	QS ₁	Status
0	0	No operation
0	1	First byte of opcode from the queue
1	0	Empty the queue
1	1	Subsequent byte from the queue

- SS2 (pin 26, 27, and 28)

These are the status signals that provide the status of operation

S ₂	S ₁	S ₀	Status
0	0	0	Interrupt acknowledgement
0	0	1	I/O Read
0	1	0	I/O Write
0	1	1	Halt
1	0	0	Opcode fetch
1	0	1	Memory read
1	1	0	Memory write
1	1	1	Passive

- **$\overline{OK}$ (pin 29)**

When this signal is active, it indicates to the other processors not to ask the CPU to leave the system bus.

- **$\overline{RQ}$ and $\overline{RG}$ (pin 30 and pin 31)**

These are the Request/Grant signals used by the other processors requesting the CPU to release the system bus. When the signal is received by CPU, then it sends acknowledgment. $\overline{RG}$ has a higher priority than $\overline{RQ}$.

- **Minimum Mode Configuration:** In minimum mode configuration, $\overline{MN}/\overline{MX}$ is connected to V_{cc}.
- In minimum mode, the pins used are $\overline{MALE}$, $\overline{ADT}/\overline{RM}/\overline{DMA}$, $\overline{HOLD}$, $\overline{HLDA}$.
- It contains a clock generator 8284 which generates clk pulses.
- Three latches 8282 are used to generate 20 bit address and $\overline{HE}$.

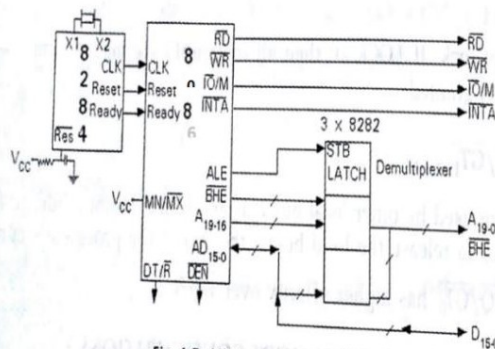


Fig. 1.7 Minimum Mode of 8086

- **MAXIMUM MODE CONFIGURATION:** In Maximum mode configuration, $\overline{MN}/\overline{MX}$ is connected to gnd.
- The signals used are $\overline{MN}/\overline{QS1}$, $\overline{QS0}$, $\overline{QS2}$, $\overline{LOCK}$, $\overline{HOLD}$ respectively.

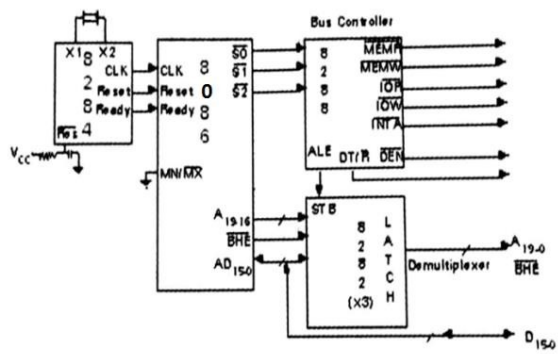


Fig. 1.8 Maximum Mode of 8086